Scenario Analysis: Issue Spotting

AI Scraping and the Future of Digital Preservation Archives

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Table of Contents

Table of Contents
Scenario and Stakes
Intended Benefits
- Possible Risks

Major Legal Questions
- Copyright Liability for facilitating AI Training
- Platform Liability and Terms of Service
- Privacy Law
- Open Access vs. Publisher Withdrawal

Resources and Research Approach
- Legal Research
- Stakeholder Engagement and Technical Consultation
- Institutional Models

Final Thoughts

Scenario and Stakes

The Internet Archive, the Library of Congress's web archiving programs, and similar digital institutions preserve the web because the web does not preserve itself. Pages disappear, URLs break, publishers revise stories, agencies remove pages, companies shut down sites, and platforms change access rules. Pew research found that a quarter of webpages that existed at some point between 2013 and 2023 were inaccessible by October 2023, and 38 percent of 2013 webpages were unavailable a decade later. Operating on a simple democratic premise that the open web should remain open for the future, these institutions work to preserve the record of the open web so that journalists, researchers, educators, courts, and future generations can know what was said, changed, deleted, or hidden. Web archives make it possible to investigate government revisions, track changes in news reporting, recover dead links, preserve community history, and resist the quiet disappearance of digital culture.

Generative AI has changed the risk profile of that mission. AI companies need enormous volumes of text, images, metadata, and historical material. Digital archives are attractive because they contain structured, time-stamped, high-value materials at scale: public domain books, OCR files, news pages, government documents, web snapshots, and records that may no longer exist on the live web. The same architecture that makes these archives valuable for preservation also makes them vulnerable to extraction.

For instance, the Internet Archive is the largest digital preservation nonprofit in the world, holding over a trillion captured webpages. Journalists use it to track editorial changes; litigants submit archived pages as evidence; historians study how political language has shifted. On Memorial Day weekend in 2023, as Brewster Kahle, the Archive's founder, documented in a public blog post, tens of thousands of requests per second for OCR text files arrived from sixty-four virtual hosts on Amazon Web Services, crashing the Archive for roughly an hour. Upon blocking those IP addresses, in just a few hours another sixty-four resumed the same behavior. Kahle wrote at the end, 'Those wanting to use our materials in bulk should start slowly, and ramp up. Also, if you are starting a large project please contact us at info@archive.org, we are here to help.' The incident shows the problem goes beyond copyright and into infrastructure, access governance, and institutional survival; revealing that the Archive's infrastructure, designed for public access, is unequipped to negotiate terms with an extractive industry operating at machine scale. An open archive can be treated as a free input pipeline by commercial actors who do not bear the costs of preservation.

The downstream consequences are now playing out in real time. According to an analysis by the AI-detection startup Originality AI, 241 news sites across nine countries now explicitly block at least one of the Internet Archive's crawling bots, with 23 major news sites blocking ia_archiverbot entirely. These include The New York Times, USA Today, and Reddit. The Guardian has adopted a more surgical approach, blocking archival access to its journalism specifically while leaving topic and homepage content available. The New York Times has gone further, implementing 'hard blocks' beyond what standard robots.txt conventions require. As Anisha Sircar reported in Forbes, USA Today Co. alone accounts for 87 percent of the blocking publishers, given its ownership of more than 200 media outlets under the former Gannett umbrella.

The stated rationale from publishers is largely uniform. The Times declared that “Times content on the Internet Archive is being used by AI companies in violation of copyright law to directly compete with us.” Reddit blocked the Wayback Machine after discovering that AI companies were using it as a free alternative to the licensing deals Reddit had negotiated with

Google and OpenAI. The Guardian's director of business affairs and licensing told Wired the outlet had been in conversation with the Archive over 'concerns over potential misuse by AI companies of content sets crawled for preservation purposes.' A 2023 analysis of Google's C4 training dataset by the Washington Post confirmed that the Internet Archive was among the sources used to build both Google's T5 model and Meta's Llama models.

This is the scenario a policy must address: a nonprofit preservation institution that is both a victim of AI extraction and, in the eyes of publishers, an accessory to it. The policy challenge is not simply to choose between open access and content protection. It is to build a framework that can distinguish between the unannounced midnight AWS blitzkrieg and the graduate student's dissertation corpus, between the commercial foundation model and the fact-checker's text-mining script, and to do so with legal precision, institutional credibility, and the technical capacity to enforce it.

Intended Benefits

The proposed policy should therefore avoid a simple “AI scraping: yes or no” answer. A blanket block would harm research, journalism, and open access. Schafer and Winters, writing in the International Journal of Digital Humanities (2021), argue that web archives require “increased access” but that “the transparency, accountability, reproducibility, sustainability and legibility of these complex knowledge infrastructures and born-digital heritage are key for their current as well as future use.” They articulate seven principles of good governance for web archives: participation, consensus, accountability, transparency, effectiveness and efficiency, inclusivity, and adherence to the rule of law. Every one of these principles is under pressure from the current situation. A policy that meets all seven will be more durable than one optimized for a single value.

But an open door invites overload, publisher retaliation, and loss of public trust. The same architectural openness that lets a researcher in Lagos read a vanished article from 2010 lets a commercial scraper hammer the servers from a cloud provider's farm. The Archive's own Vanishing Culture report (Messarra, Freeland, and Ziskina, Internet Archive Press, 2026) frames the resulting harm as a digital memory hole, where the cumulative effect of corporate licensing, platform consolidation, publisher withdrawal, and direct cyberattack is the slow erasure of the public's ability to construct and access its own cultural record. The proposed policy is meant to do four things at once: protect the Archive from operational collapse; clarify its legal exposure when third parties exploit its collections to train models; establish enforceable distinctions between scholarly and commercial use, since the technology cannot draw that line itself; and head off the publisher withdrawal that is already shrinking the historical record, as the New York Times, Reddit, and most of USA Today's two hundred outlets have begun blocking the Archive's crawlers to keep AI companies away from their work.

Possible Risks

The policy must make deep considerations towards various factors, as it can create risks if not foreseen well. For instance, credentialing can exclude independent researchers, journalists, local historians, community archivists, and students without institutional backing. A researcher passport must therefore include non-university pathways. Tiered access can become slow and bureaucratic. Sometimes a dissertation project may look like a bot and technical controls could overblock. The policy needs appeal channels and human review.

Machine-readable preference signals may be ignored by bad actors. Robots.txt and TDMRep are useful signals, but not complete barriers. A more complete policy might need logs, rate limits, contractual terms, and enforcement. Reciprocity requirements may favor large AI companies that can negotiate agreements and burden small public-interest projects. The policy should thus scale obligations to risk and capacity as well.

Major Legal Questions

Copyright Liability for facilitating AI Training

The foundational legal question is whether AI companies that ingest archived web content for model training infringe the copyright of the original publishers whose work appears in those archives. This question is actively litigated in more than one hundred pending cases in the United States. The governing framework is Section 107 of the Copyright Act, which provides a four-factor fair use defense based on the purpose and character of the use, the nature of the copyrighted work, the amount used, and the effect on the market for the original.

The Library Copyright Alliance has argued, consistent with the reasoning in *Authors Guild v. HathiTrust*, 755 F.3d 87 (2d Cir. 2014), that ingestion for training purposes can be "quintessentially transformative" because a model trained on text does not reproduce that text for reading purposes; it uses it to learn statistical relationships. The HathiTrust court found that creating a full-text searchable database from millions of books was transformative even though it involved copying entire works, because the output served a different function from reading the original. *Authors Guild v. Google* (2d Cir. 2015) extended this reasoning to snippet display. *Bartz v. Anthropic* and *Kadrey v. Meta* (both N.D. Cal. 2025) addressed the transformative nature of training on copyrighted books. These cases gave the library and research community confidence that text and data mining for scholarly purposes does not infringe. The U.S. Copyright Office, in its May 2025 report on generative AI training (Part 3), drew on this precedent while noting the important limiting principle: noncommercial research and analysis uses are 'likely to be deemed fair,' but commercial uses producing outputs that compete with the copyrighted works used in training are considerably more difficult to defend.

Thomson Reuters Enterprise Centre v. Ross Intelligence, 765 F. Supp. 3d 382 (D. Del. 2025) provided that limiting principle in practice. The court found that an AI company that trained a legal research tool on West headnotes could lose the fair use defense where the tool's outputs function as a market substitute for the original. That reasoning maps directly onto AI companies training on archived journalism and then offering news summary services that compete with the publishers whose work they ingested. The policy should therefore separate low-risk computational research from high-risk commercial model training that may compete with the works or collections being used.

Platform Liability and Terms of Service

Can the Internet Archive be held legally responsible for how third parties exploit access to its collections? This question sits at the intersection of copyright law and the intermediary liability framework established by Section 512 of the DMCA and, historically, Section 230 of the Communications Decency Act. DMCA §512 may protect service providers in some circumstances, especially for user-uploaded content, but it will not necessarily immunize an archive’s own reproduction, indexing, or access design. Section 230 may protect against certain claims involving third-party content, but it does not apply to federal intellectual property claims. The Archive has relied on its DMCA safe harbor to respond to take-down requests on a reactive basis rather than proactively policing all uses. But the scale of AI extraction challenges that posture.

Lemley and Henderson have argued in *The Mirage of Artificial Intelligence Terms of Use Restrictions* that most AI-related ToS provisions are unenforceable in practice because they cannot bind anonymous bulk scrapers who never see the terms. *ProCD, Inc. v. Zeidenberg*, 86 F.3d 1447 (7th Cir. 1996), and *Doe v. GitHub* (currently on appeal in the Ninth Circuit) mark the bounds of what can be done with click-through and registration terms.

Section 1201 protects technological measures that control access to copyrighted works. The Archive currently operates an open public Scraping API at archive.org/services/search/v1/scrape, with no terms restricting commercial AI use. If the

Archive deploys API authentication and rate controls as access conditions, circumvention of those measures, including the kind of automated retry logic that commercial scraping vendors openly market, becomes a separate statutory violation. Section 230 of the Communications Decency Act, 47 U.S.C. § 230, has historically protected hosts from liability for third-party content. Section 1202 prohibits the removal or alteration of copyright management information. Several plaintiffs in the current generation of AI cases have argued that AI training pipelines strip authorship metadata in ways that violate Section 1202. The 1201 rulemaking exemptions for nonprofit text and data mining must be respected so that legitimate scholarly research is not blocked along with commercial extraction.

Privacy Law

Web archives may include personal data, deleted posts, sensitive community records, government pages, social media, etc. Home addresses on civic websites, health information on archived forum posts, immigration status in court records, identity data from social media profiles that users may have subsequently deleted, all frequently fall into archived pages. Even when material was once public, preserving and extracting it at scale may create new risks. Reddit's concerns about archived user comments and profiles illustrate this problem. European data protection law, and increasingly state privacy statutes in the United States, may create obligations for the Archive to honor deletion requests for personal data even within its historical collections.

Zannettou et al. (2018), studying the use of web archiving services on social media, noted that archiving services 'serve a crucial role in ensuring that Web content persists, but do so without regard to (and often in spite of) the rights and consent of content producers.' This concern is especially acute when AI training amplifies the reach of archived personal data, since a model trained on archived content may reproduce or generate content that affects real individuals who never consented to their information being used in this way. The policy may thus need to distinguish between public-interest preservation and high-volume reuse that increases privacy risk.

The NASCIO-CoSA State Archiving Playbook (2018) provides relevant guidance here, noting that records managers must distinguish between records requiring long-term preservation and records that should be deleted once their retention period expires. Its principle that improved records management 'can also reduce legal risk associated with non-compliance with state or federal laws and regulations' can be argued to apply with equal force to privacy obligations in the web archiving context. For sensitive collections, the policy should borrow from restricted-data models such as ICPSR, data management plans, institutional review, and researcher agreements. It should also consider the CARE Principles for Indigenous Data Governance where community data is involved. Open access is a value, but it is not the only value. Preservation must be balanced with privacy, safety, donor intent, and community control.

As Lerman et al. (arXiv, February 2026) documented in work on large-scale online deanonymization with LLMs, AI tools can now re-identify individuals across datasets from materials that were believed to be sufficiently anonymized. The Archive's policy must therefore treat de-identification not as a single self-executing standard but as a documented process subject to ongoing review, consistent with HHS guidance on the HIPAA de-identification framework and with the NASCIO-CoSA State Archiving Playbook's principle that records should be “appropriately identified, properly classified, and maintained according to retention policies.”

Open Access vs. Publisher Withdrawal

Perhaps the most consequential legal and policy question is not what AI companies may do, but what happens if publishers, in response to AI scraping, withdraw from the archival ecosystem entirely. Kate Knibbs's reporting in Wired documents the irony precisely: USA Today Co. recently published an investigation into ICE detention statistics that relied heavily on Wayback Machine data, while simultaneously blocking the Archive from crawling its own journalism. Wayback Machine director Mark Graham stated in response, “there's no question that the general locking-down of more and more of the public web is impacting society's ability to understand what's going on in our world.”

If the trend continues, the historical record of early twenty-first century digital journalism will not exist in twenty years. Legal proceedings, investigative reporting, and democratic accountability all depend on the Archive's ability to document what institutions said and when they changed their story. As the Electronic Frontier Foundation's Joe Mullin has argued, blocking the Archive is 'like a newspaper publisher announcing it will no longer allow libraries to keep copies of its paper.' The Archive's policy must give publishers a viable alternative to hard blocking, with something that provides a credible assurance that bulk access to archived versions of their content will be governed by enforceable terms that prohibit commercial AI training without consent.

The Computer Fraud and Abuse Act (CFAA), 18 U.S.C. § 1030, has been the subject of two important recent decisions. *hiQ Labs v. LinkedIn* (9th Cir. 2022) narrowed the statute considerably for ordinary scraping of public data. *X Corp. v. Bright Data* (N.D. Cal. 2024) reinforced that a platform cannot use the CFAA to convert public data into private property. hiQ held that LinkedIn could not use the Computer Fraud and Abuse Act to block a data analytics company from scraping publicly available profile data. Bright Data extended this reasoning, finding that sweeping ToS restrictions on data extraction cannot “yank public information into a private domain.” If publishers and platforms can block archives entirely, the public record becomes dependent on private permission, creating information asymmetry. Both cases suggest that giving any single party, including a nonprofit, unlimited power to restrict access to public data carries serious risks for the information commons.

At the same time, publishers have legitimate interests. They invest in journalism, face commercial pressure, and object to AI firms using archived copies to avoid licensing. The policy must therefore respect both sides. It should not treat all publisher restrictions as bad faith. But it should argue that hard blocking preservation institutions is a poor solution because it sacrifices accountability journalism, historical research, and public memory to fight a separate battle with AI companies. As the framework of “good governance for web archives” provided in the International Journal of Digital Humanities (2021) shows us: a policy that responds to AI threats by restricting all access fails the inclusiveness criterion, while a policy that ignores publisher concerns fails the accountability criterion.

This is the point at which the argument from the scholarly communications literature becomes most useful. As the Open Access and AI training rights framing argues, “the problem is not AI training on open archives per se. The problem is uncredentialed, unaccountable, unlimited extraction with no transparency, no reciprocity, and no consent mechanism.” A policy built on conditionality and reciprocity rather than categorical restriction addresses the publisher's actual concern, protects legitimate research, and preserves the Archive's mission.

Resources and Research Approach

Legal Research

Primary legal research should begin with examination of the the four-factor fair use framework as applied in Authors Guild v. HathiTrust (755 F.3d 87 (2d Cir. 2014)) and Authors Guild v. Google (804 F.3d 202 (2d Cir. 2015)), both of which establish that large-scale digital copying for transformative purposes can be fair use. Thomson Reuters v. Ross Intelligence (D. Del. 2025) should be read as the limiting case. Bartz v. Anthropic is instructive on the distinction between AI input issues and AI output issues. The U.S. Copyright Office's May 2025 report on generative AI training (Part 3) is an essential secondary source, particularly its analysis of market harm under the fourth factor and its observation that “the copying involved in AI training threatens significant potential harm to the market for or value of copyrighted works." Section 1202 CMI removal doctrine, as developing in Kadrey v. Meta and related proceedings, warrants separate attention as a possible independent basis for liability.

Secondary legal research should canvass the Library Copyright Alliance's published principles on AI and libraries, the Vienna Principles for Scholarly Communication, the Electronic Frontier Foundation's analysis of the publisher blocking trend, and the emerging W3C Text and Data Mining (TDM) standards that would allow machine-readable opt-out signals. The European Union's copyright directive already recognizes machine-readable TDM opt-outs as legally significant, and U.S. courts may take notice of that framework as they develop the American fair use doctrine in the AI context.

Stakeholder Engagement and Technical Consultation

Stakeholder engagement should include practicing archivists (particularly those at institutions that have signed onto the UVA Archival AI Protocol, as documented by Kelley Slebos's work in the archival AI best practices literature), journalists who have used the Wayback Machine for investigative work, legal counsel familiar with DMCA safe harbor litigation, publishers who have implemented blocks, academic researchers conducting computational analysis on archived collections, and representatives of AI companies that have sought bulk access. I plan to contact archival staff who manage the Wayback Machine's crawling infrastructure as well, determining how the institution is currently thinking about these questions. The journalist coalition letter organized by the EFF, Fight for the Future, and Public Knowledge, signed by more than one hundred reporters, also provides a natural entry point for journalism stakeholder contact.

Technical consultation should address how more local libraries and similar institutions manage mass scraping of their materials on their digital front. I have already done preliminary site visits to NC State Libraries and the State Library of North Carolina, where staff explained how their organizations use traffic prioritization and rate-limiting to protect against bulk extraction. Those conversations will inform the operational sections of the policy. Technical research should also include the Internet Archive's API documentation, robots.txt, proposed ai.txt or similar preference signals, bot detection, tiered quality-of-service prioritization, cloud traffic monitoring, and researcher credentialing.

Institutional Models

Further, I will be comparing different strategies employed by different institutions. The University of Virginia Archival AI Protocol, published in January 2026 by Leo S. Lo, offers a template that other archival organizations are beginning to adopt. The Harvard Library's release of a public domain training corpus in late 2025 offers a different model with libraries as active providers of curated training data rather than passive resistors. Both will inform the final memo, even where the Archive's situation requires departures from each.

Final Thoughts

Once the three research tracks have produced a working map of legal requirements, stakeholder needs, and technical capabilities, the policy can move to drafting. At that stage the central work is distinguishing between what the law requires, what good institutional stewardship requires beyond legal minimums, and what is aspirational but not yet enforceable.

The analysis should be tested against five concrete scenarios: a journalist using archived pages for fact-checking; a graduate student running text mining on a sample of archived news for a dissertation; a university research team conducting large-scale NLP analysis of archived government pages; a nonprofit AI safety lab building a misinformation detection model using archived journalism; and a commercial LLM company seeking bulk extraction of the full Wayback Machine for a general-purpose AI product. If the draft policy treats all five identically, it is not yet a policy. The goal is a framework precise enough to reach different conclusions for these different situations while remaining simple enough to administer with the finite resources preservation institutions actually have.

The best final policy will be short, enforceable, and public-facing. It will most likely declare that open access remains the default for human use and public-interest preservation; bulk access must be transparent and proportionate, sensitive data receives added review, commercial training requires agreement, provenance must be preserved, and those who extract value at scale must return value to the public infrastructure that made the work possible.